Qualifying the Measurement Instrument

Phase 0 of the Phase 1 replacement programme — and five occasions on which a measurement artefact was read as a result about the method

August 2026

Abstract

Before approaches **B** (auction-dynamics MBO) and **C** (capacity-constrained geodesic Lloyd) can be compared against the PGD control on perimeter, the things that would do the comparing must themselves be shown to work: a *balanced-assignment solver* shared by all three approaches, and a single *evaluation harness.* This report records that qualification. Four results. (1) The harness reproduces the validated N=100 deliverable's Phase 2 perimeter **bitwise** — 185.2546144718457 — from a completely different entry path, which simultaneously establishes that hardening an arm's labels costs it nothing and that Phase 2 is bit-deterministic on this host. (2) The shared solver needed *no new algorithm*: the conclusion that an auction or Sinkhorn solver was required came from a fixture modelling a state the arms occupy for one outer iteration. (3) Under the final protocol (three meshes × two arms × three seeds) the solver passes **18/18**, and the gate is shown to discriminate — the pre-0a un-normalized solver is *indistinguishable from doing nothing.* (4) Measured at genuinely equal footing the arms sit at **parity,** ±10%, with the incumbent, not clear of it; two earlier versions of this claim were wrong in the arms' favour. The methodological content is larger than the technical content: **five** times during this work a defect in the measurement was read as a result about the method, and the report documents what caught each one.

Provenance

Status	**measured**
Date	2026-08-16 to 2026-08-18
Surface	torus $T(R{=}1.0,\ r{=}0.6)$ throughout
Fixtures	V=47,488 (224×212) at N=300; V=114,144 (348×328) at N=100 and at N=300 (the production configuration). Seeds $0, 1, 2$.
Control	`results/run_20260709_081548_surftorus_npart100_...._lam5.1_seed84172851`, the validated N=100 deliverable: Phase 2 best perimeter **185.2546144718457** at iteration 20 of 20, campaign `ipopt_btol0.001_lbfgs30_hess_bestiter_partial`, worst cell 0.7786%, 0 dead / 0 weak / 0 fragmented.
N=300 baselines	`run_20260806_123326` (V=47,488, Phase 2 best 323.319247087254, plateaued at iterate 18 of 19) and `run_20260808_191030` (V=114,144, 322.96218780465847, *censored* — best is its final iterate). These are *one* experiment: the second resumed from the first (`warm_start_path`). Both were copied into this checkout on 2026-08-18 (12.6 GB, verified identical).
Incumbent	approach A's own dual stage on its own log densities, normalized, at the same 2000-iteration budget the arms are scored with: **1.3517**% at V=47,488, **0.5991**% at V=114,144, both N=300.
Scripts	`testing/test_balanced_assignment_solver.py` (gates 1–4); `testing/test_arm_harness.py` (0b acceptance); `testing/test_phase0d_shakedown.py` (0d, ∼34 min); `docs/experiments/07-phase0-shared-harness/make_figures.py`.
Code	`src/partition/balanced_readout.py` (`solve_dual_offsets`, `assignment_margin_scale`); `src/partition/arm_harness.py` (`assignment_quality_bar`, `A_PARITY_NORMALIZED`).
Versions	Python 3.12.13, numpy 2.4.4, scipy 1.17.1, h5py 3.16.0; macOS 15.3.1 arm64 (Mac mini).
Anchors	0d reproduces 185.2546144718457 **bitwise**; final gate 2 is **18/18** in 2 h 39 m; the null solver scores 16.588% at the production pin against a 0.841% bar.

1 Where B and C come from

Neither arm is invented here; both are established constructions applied to this surface and this pipeline. This section states the provenance explicitly, because an earlier draft of this report carried none.

B — volume-constrained threshold dynamics. Threshold dynamics originates with Merriman, Bence and Osher [14]: alternate short-time diffusion of the phase indicators with a pointwise reassignment, and the interfaces move by (approximate) mean curvature. Esedoğlu and Otto [7] put the multiphase case on a variational footing, exhibiting a Lyapunov functional that Γ-converges to a weighted perimeter — the same limit our own energy targets [3, 15]. The specific variant we would build is Jacobs, Kim and Léger's *auction dynamics* [10], which imposes exact volume constraints by replacing the pointwise threshold with a *balanced assignment*, solved by Bertsekas' auction algorithm [2]. That paper is the direct antecedent: it is volume-constrained multiphase MBO, published for this problem class.

Two caveats the literature itself supplies. Connectivity is *not* guaranteed per step — stray components shrink under curvature flow but are not excluded — and on graphs and coarse meshes threshold dynamics is known to *freeze* when the diffusion length falls to the discretization scale [16]. Both shaped the design: the first sets the validity grading, the second fixes $\tau = (4h)^2$ rather than h^2.

C — capacity-constrained centroidal power diagrams. Lloyd's algorithm [12] iterates "assign to nearest site, move site to centroid". Adding a prescribed-capacity constraint turns the assignment into a semi-discrete optimal transport problem whose solution is a *power diagram* with per-site weights — the Minkowski-type result of Aurenhammer, Hoffmann and Aronov [1]. Mérigot [13] and Kitagawa, Mérigot and Thibert [11] give the modern solvers for those weights; de Goes *et al.* [6] use exactly this construction for blue-noise sampling. On a curved surface the Euclidean distance is replaced by a geodesic one, for which the heat method [4] is the standard alternative to the graph-Dijkstra approximation used in our fixtures.

C minimizes a *quantization* energy, not perimeter. It lands near the perimeter optimum only because both optima are asymptotically hexagonal in the plane — Hales for perimeter [9], Fejes Tóth and Gersho [8] for quantization. **Both are flat-plane results**, and this torus has Gaussian curvature of both signs, so neither transfers; how far C lands from optimal is precisely what remains to be measured.

The shared solver, and what was nearly rebuilt. All three approaches need the same primitive, and it is the semi-discrete OT dual [1, 13]: find per-cell offsets ψ such that $\arg\max_k [\, s_{ik} + \psi_k \,]$ has equal masses. The shipped implementation is subgradient ascent on that dual. Section 4 records that this report's programme nearly replaced it with an auction [2] or an entropic (Sinkhorn) solver [5] — both standard, either defensible — on evidence that turned out to be a fixture artefact. Sinkhorn was implemented and measured; it *lost* to the incumbent on five of six score matrices.

2 Why Phase 0 exists

The Phase 1 relaxation costs **22.0–79.3 h** for a completed N=300 solution and its winner-take-all readout is not a valid partition, so it is repaired afterwards. That does not scale. Approaches B and C would produce a valid partition at every step in minutes. Deciding between them requires measuring *perimeter after Phase 2*, and both — plus the shipped approach A — need one shared primitive: *balanced assignment given arbitrary per-vertex scores*, which exists inside the balanced readout as `solve_dual_offsets`.

Phase 0 makes that primitive usable on foreign score scales (0a), builds one evaluation harness (0b), adopts the N=300 baselines (0c), and proves the harness reproduces a known answer (0d). **It measures nothing about perimeter**; every question about whether B or C is any good remains open.

3 0d — the harness reproduces a known answer, bitwise

The strongest single result. The N=100 control's own winner-take-all labels were hardened to one-hot densities, written in the Phase 1 solution schema, and pushed through the entire harness under the control's own campaign `refinement.yaml`:

reference	185.2546144718457		
harness	185.2546144718457	(iterate 20 of 20)	
relative difference	0.000000%	bitwise: **true**	wall 2023 s

Independently verified: the scratch campaign holds 20 fresh iteration files, its `refinement.yaml` names the one-hot arm file as `base_solution`, and the *entire* 20-iterate trajectory is bitwise equal to the control campaign computed five weeks earlier, including the 14,883-entry `lambda_parameters` array.

Three consequences. **(a)** The apples-to-apples question is settled exactly: an arm's hard labels and PGD's continuous field produce identical Phase 2 initial states, because `ContourAnalyzer` reads densities only through `argmax` and every variable point starts at λ=0.5. **(b)** Phase 2 is bit-deterministic on this host, previously an unverified load-bearing assumption. **(c)** The control is itself *censored* — its best iterate is its last — so 185.2546144718457 is a *bound*.

4 0a — no new solver was needed

`solve_dual_offsets` is subgradient ascent on the semi-discrete OT dual. It is *correct* for any additive score matrix but its step schedule was calibrated to the log-density scale. Normalizing scores by a robust assignment-margin scale, and raising the step (`normalized_eta0` 0.5 $\to$ 2.0; measured 3.654 $\to$ 2.209 $\to$ **1.672** $\to$ 2.248% for $\eta_0 = 0.5, 1, 2, 4$) and the budget (400 $\to$ 2000) is the whole fix. Approach A's path is untouched and byte-identical.

4.1 The conclusion that was withdrawn

The first version of gate 2 scored a *cold start* — balance a raw geometric guess in one shot — and B failed at 45.35%. This was written up as structural: the shipped solver is subgradient ascent tuned for a small correction, B and C need to rebalance ~230 of 300 cells, therefore build a Bertsekas auction or an entropic solver. **All of that was wrong.** B and C are *iterative*: from outer iteration 1 each assignment starts from an already-balanced partition. An iterated Lloyd loop with the *existing* solver runs 23 consecutive iterations under the bar. Sinkhorn, implemented and measured, *loses* to the incumbent on 5 of 6 score matrices. Two knobs dismissed as dead — budget and step size — each fix it on their own; the "a 10× budget costs 20 minutes" figure was operations *counted*, never timed (the true cost is 33.9 ms/iteration).

5 The gate, and whether it discriminates

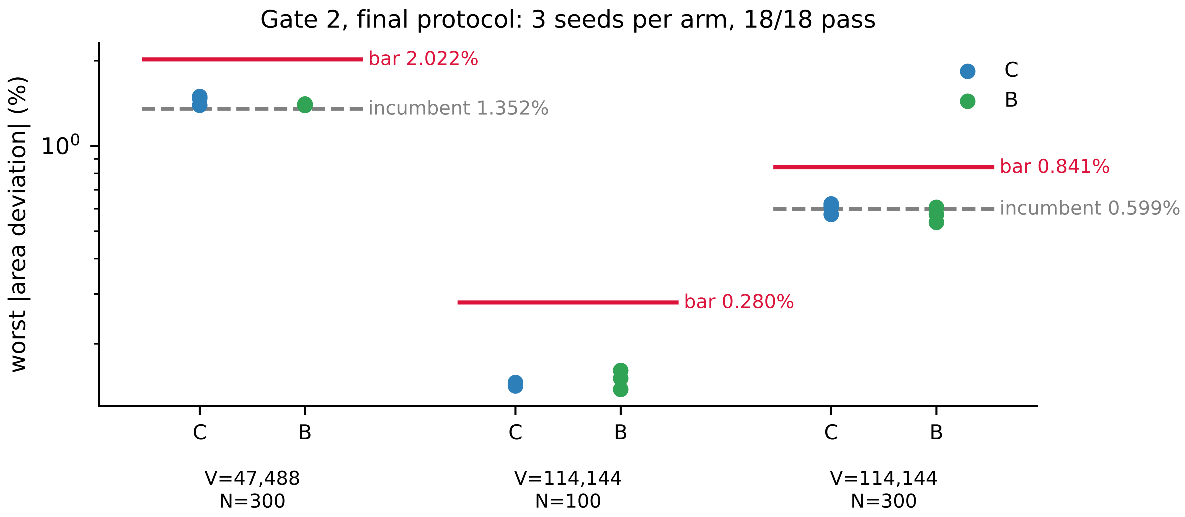

Figure 1: Final protocol: three fixtures, two arms, three seeds, 18/18 pass. Red line: the acceptance bar. Grey dashed: the incumbent at equal budget.

The bar is max(2 × vertex granularity, A's own dual stall at equal budget). A bar the shipped solver itself fails cannot be evidence about a foreign score scale; a bar below the one-vertex granularity floor is impossible by construction.

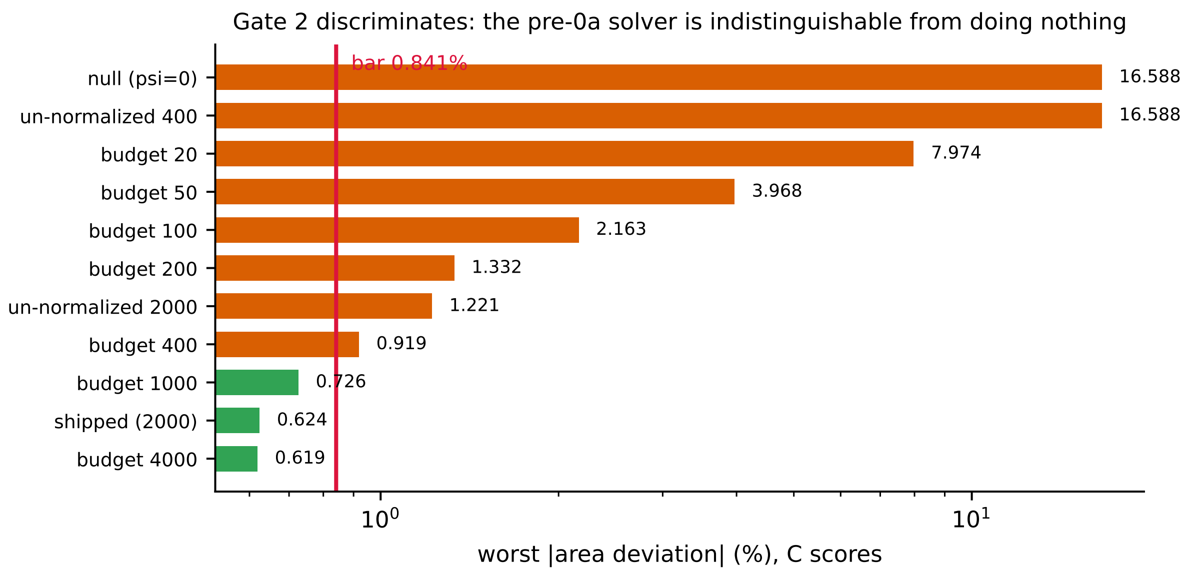

Figure 2: Degraded solvers at the production pin, C scores. The pre-0a un-normalized solver scores 16.588% — *equal to the null solver to three decimals*, because its best iterate is its ψ=0 starting point.

The shipped configuration sits 27–48% under its bars with the failure boundary a factor 2–5 away. Gate 3 asserts a null solver *fails*; gate 4 verifies that early stopping is safe inside an iterated loop (3.0× faster for +0.008 points of degradation).

6 The result, stated at equal footing

fixture	bar	C (s0/s1/s2)	B (s0/s1/s2)
V=47,488, N=300	2.022%	1.472 / 1.392 / 1.495	1.405 / 1.390 / 1.406
V=114,144, N=100	0.280%	0.146 / 0.142 / 0.144	0.161 / 0.151 / 0.138
V=114,144, N=300	0.841%	0.624 / 0.573 / 0.609	0.537 / 0.573 / 0.607

Against the incumbent's 0.5991% at the production pin, C spans 0.573–0.624% and B spans 0.537–0.607%: **parity,** ±10%. Earlier drafts of this claim said "2–3× better" and then "clear of parity by 30%"; both compared arms at 2000 iterations against A at 400. *A budget difference wearing a method difference's clothes.* The assignment step is not a differentiator and was never meant to be one — 0a's job was to make the shared primitive usable so that perimeter can be measured.

7 Five measurement artefacts read as method results

1. **The stall guard** warned on any floored line search — which is how a healthy level normally ends. Two commits reasoned from the false positive.
2. **A corpus glob** reached 130 of 350 traces, and the resulting "empty gap" became a report's strongest argument.
3. **A baseline audit** enumerated one working directory and concluded no N=300 Phase 2 baseline existed. Two did, in a sibling worktree.

4. **The theatre bar**: gate 2's bar included 1.25× a strong reference computed *with the solver under test*, so a broken solver inflated its own bar. Six crippled solvers passed, the weakest doing nothing at all.

5. **The early-stop mask**: with early stopping inside the measurement path, gate 2 passed 12/12 while every number hugged its bar from below (Figure 3).

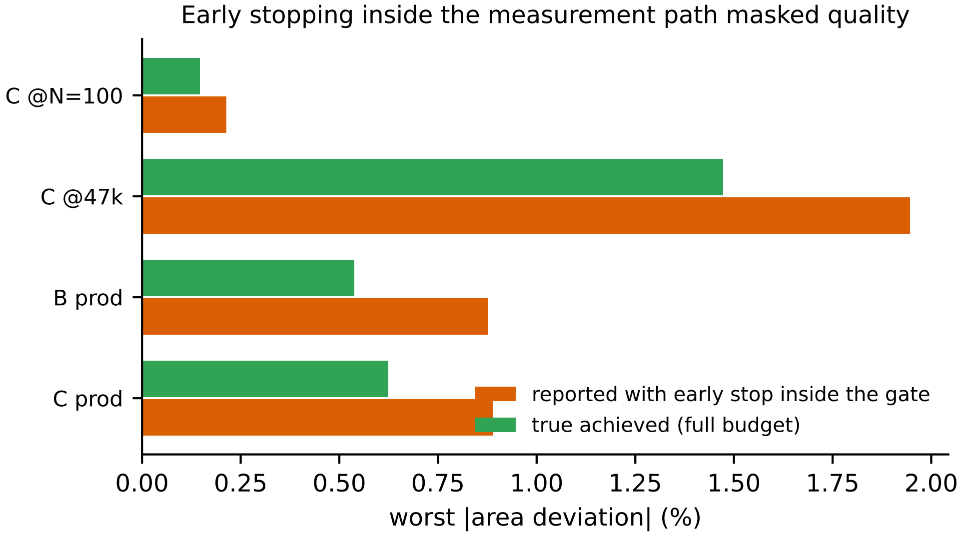

Figure 3: Artefact 5. A passing gate looks identical either way; only the pre-registered prediction distinguished them.

Two things caught these, and neither was insight. **Pre-registration**: the early-stop mask surfaced only because predicted values were committed before the run, and four "missed" in a suspiciously uniform direction. **Asking whether anything fails**: every test asked *"does the shipped solver pass?"* and none asked *"does anything fail?"*, so a gate that passed everything looked exactly like a gate that worked. Gate 3 now closes that.

On "survived adversarial review". Four adversarial reviews inform this work and each found real defects. They were subagents dispatched *from the same session as the work they reviewed* — independent of the author's reasoning, not of the author's framing, tooling, or choice of what to point them at. Every load-bearing number they produced has since been self-reproduced, except one marked indicative in the plan. That caveat belongs next to the claim.

8 Conclusions

1. The harness is qualified: it returns a known answer bitwise, and hardening an arm's labels costs nothing measurable.

2. No new assignment solver is needed. Normalization, step size and budget suffice; the auction/Sinkhorn conclusion was a fixture artefact.

3. The gate discriminates: 18/18 for the shipped configuration, with the pre-0a solver indistinguishable from doing nothing.

4. Arms and incumbent are at parity ±10% on assignment quality. The open question — whether B or C produces *shorter perimeter* — is untouched by this report.

Related documents

- Plan and pre-registration: `docs/plans/PHASE1_BC_REPLACEMENT_PLAN.md`
- Taxonomy: `docs/reference/PHASE1_HIGHN_APPROACHES_ABCDE.md`
- The gap being closed: `docs/reference/winner_take_all_partition_gap.md`
- Prior arms: `docs/experiments/05-soft-area-constraint/`, `docs/experiments/06-subfloor-ladder/`

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